Outline

1. Optimization Model: Standard Form and Keywords
   - Standard Form
   - Optimal and Locally Optimal Points
   - Equivalent Optimization Problems
   - Quasiconvex Optimization
2. Optimality Conditions

Optimization Problems

Optimization problem (Standard Form):

$$\begin{aligned} \text{minimize} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, i = 1, \ldots, m \\ & h_i(x) = 0, i = 1, \ldots, p \end{aligned}$$

- Find an x that minimizes $f_0(x)$ among all x that satisfy
  - $f_i(x) \leq 0,\ i = 1, \ldots, m$
  - $h_i(x) = 0,\ i = 1, \ldots, p$
- x is called optimization variable
- $f_0 : \mathbb{R}^n \to \mathbb{R}$ is called objective function or cost function
- The inequalities $f_i(x) \leq 0$ are called inequality constraints
- $f_i : \mathbb{R}^n \to \mathbb{R}$ are called inequality constraint functions
- The equations $h_i(x)$ are called equality constraints
- The functions $h_i : \mathbb{R}^n \to \mathbb{R}$ are called equality constraint functions

Optimization Problems

Optimization problem (Standard Form):

$$\begin{aligned} \text{minimize} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, i = 1, \ldots, m \\ & h_i(x) = 0, i = 1, \ldots, p \end{aligned}$$

- If no constraints, i.e., $m = p = 0$, then called unconstrained problem
- Domain of opt. problem: where objective and constraint are defined

$$\mathcal{D} = \bigcap_{i=0}^{m} \mathbf{dom}\ f_i \quad \cap \quad \bigcap_{i=1}^{p} \mathbf{dom}\ h_i$$

- A point x is called feasible if it satisfies the constraints

$$\begin{aligned} f_i(x) \leq 0, \quad & i = 1, \ldots, m \\ h_i(x) = 0, \quad & i = 1, \ldots, p \end{aligned}$$

- The optimization problem is called feasible if there exists at-least one feasible point

Optimization Problems

Optimization problem (Standard Form):

$$\begin{aligned} \text{minimize} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, i = 1, \ldots, m \\ & h_i(x) = 0, i = 1, \ldots, p \end{aligned}$$

- Feasible set or constraint set: set of all feasible points
- Optimal Value: The optimal value p^* defined as

$$p^* = \inf \{ f_0(x) \mid f_i(x) \leq 0,\ i = 1, \ldots, m,\ h_i(x) = 0,\ i = 1, \ldots, p \}$$

- p^* is allowed to take extended values $\pm\infty$
- Infeasible problem: problem is called infeasible when $p^* = \infty$
  - Note: we used the fact that $\inf \phi = \infty$
- Unbounded below: Problem is unbounded below if $f_0(x_k) \to -\infty$ as $k \to \infty$

Optimal and locally optimal

Optimization problem (Standard Form):

$$\begin{aligned} \text{minimize} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, i = 1, \ldots, m \\ & h_i(x) = 0, i = 1, \ldots, p \end{aligned}$$

- Optimal Point: x^* is called optimal if it solves the given optimization problem, i.e.,
  - x^* is feasible point
  - $f(x^*) = p*$, that is, at x^* optimal value p^* is obtained
- Optimal Set: The set of all optimal points is called optimal set
- If there exists an optimal point, then we say that optimal value is achieved and the problem is solvable
- If the optimal set is empty, then we say that optimal value is not attained

Optimal and Locally Optimal Points

Optimization problem (Standard Form):

$$\begin{aligned} \text{minimize} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, i = 1, \ldots, m \\ & h_i(x) = 0, i = 1, \ldots, p \end{aligned}$$

- $\epsilon-$suboptimal: $\bar{x}$ is $\epsilon-$suboptimal if $f(\bar{x}) \leq p^* + \epsilon$, $\epsilon > 0$
  - $\bar{x}$ is just ϵ more than optimal value p^*
- $\epsilon-$suboptimal set: set of all $\epsilon-$suboptimal points
- Active constraint: If x is feasible and $f_i(x) = 0$, then ith inequality is active
- Inactive constraint: If x is feasible and $f_i(x) < 0$, then this constraint is inactive
- Redundant constraint: A constraint is redundant if removing it does not change the feasible set

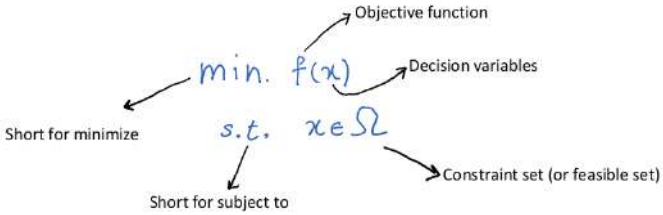
Optimization problem (Standard Form):

$$\begin{aligned} \text{minimize} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, i = 1, \ldots, m \\ & h_i(x) = 0, i = 1, \ldots, p \end{aligned}$$

Define the feasible set

$$\Omega = \{x \mid f_i(x) \leq 0, \quad i = 1, \ldots, m, \quad h_i(x) = 0 \quad i = 1, \ldots, p\}$$

More compactly, we we can write:



Examples: $1/x$

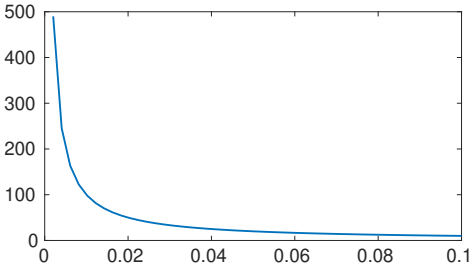
Consider the optimization problem:

$$\begin{aligned} &\text{minimize } f_0(x) = 1/x, \\ &\text{subject to } x \in \mathbb{R} \end{aligned}$$

where $f_0 : \mathbb{R}_{++} \to \mathbb{R}$

Quiz: What is feasible set?
Quiz: What is p^*?
Quiz: Is the optimal value achieved?

Figure: Plot of $1/x$



Examples: $-\log x$

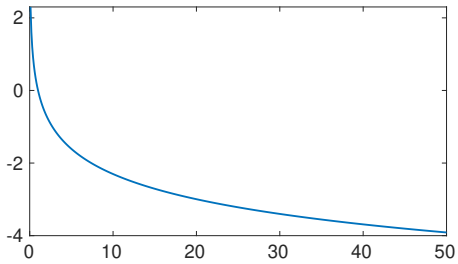
Consider the optimization problem:

$$\begin{aligned} \text{minimize } & f_0(x) = -\log x, \\ \text{subject to } & x \in \mathbb{R} \end{aligned}$$

where $f_0 : \mathbb{R}_{++} \to \mathbb{R}$

Quiz: What is feasible set?
Quiz: What is p^*?
Quiz: Is the optimal value achieved?
Quiz: Is this problem bounded below?

Figure: Plot of $-\log x$



Examples: $x \log x$

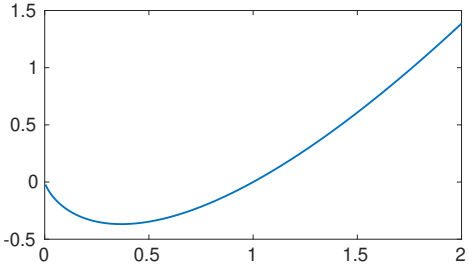
Consider the optimization problem:

$$\begin{aligned} \text{minimize } & f_0(x) = x \log x, \\ \text{subject to } & x \in \mathbb{R} \end{aligned}$$

where $f_0 : \mathbb{R}_{++} \to \mathbb{R}$

Quiz: What is feasible set?
Quiz: What is p^*?
Quiz: Is the optimal value acheived?
Quiz: Is this problem bounded below?
Quiz: What is optimal point?

Figure: Plot of $x \log x$



Expressing Problems in Standard Form

Optimization problem (Standard Form):

$$\begin{aligned} \text{minimize} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p \end{aligned}$$

Convention for standard form:

- right-hand side of the inequality and equality constraints are zero
  - For example, $g_i(x) = \tilde{g}_i(x)$ can be written as $h_i(x) = 0$
- $f_i(x) \geq 0$ as $-f_i(x) \leq 0$

(Box Constraints). Consider the following

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & l_i \leq x_i \leq u_i, \quad i = 1, \ldots, n \end{array}$$

- constraints here are called variable bounds or box constraints

The problem above can be expressed in standard form:

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & l_i - x_i \leq 0, \quad i = 1, \ldots, n \\ & x_i - u_i \leq 0, \quad i = 1, \ldots, n \end{array}$$

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, n \end{array}$$

where $f_i(x) = l_i - x_i,\ i = 1, \ldots, n$ and $f_i(x) = x_{i-n} - u_{i-n},\ i = n+1, \ldots, 2n$

Maximization Problems Seen as Minimization Problems

Note: Maximization problem can be solved by minimization. Consider

$$\begin{array}{rll} \text{maximize} & f_0(x) & \\ \text{subject to} & f_i(x) \leq 0, & i = 1, \ldots, m \\ & h_i(x) = 0, & i = 1, \ldots, p, \end{array}$$

It can be solved by solving a corresponding minimization problem with $-f_0(x)$

Optimization problem (Standard Form):

$$\begin{array}{rl} \text{minimize} & -f_0(x) \\ \text{subject to} & f_i(x) \leq 0, i = 1, \ldots, m \\ & h_i(x) = 0, i = 1, \ldots, p \end{array}$$

- Note the constraints remain same!
- Obviously, the optimal value p^* is

$$p^* = \sup \{ f_0(x) \mid \quad f_i(x) \leq 0, \ i = 1, \ldots, m, \quad h_i(x) = 0, \ i = 1, \ldots, p \}$$

Equivalent Optimization Problems

Two problems are equivalent if from a solution of the one, a solution of the other can be found

Example: Consider the following problem

$$\begin{aligned} \text{minimize} \quad & \alpha_0 f_0(x) \\ \text{subject to} \quad & \alpha_i f_i(x) \leq 0, \ i = 1, \ldots, m \\ & \beta_i h_i(x) = 0, \ i = 1, \ldots, p \end{aligned}$$

where $\alpha_i > 0, \ i = 0, \ldots, m, \quad \beta_i \neq 0, \ i = 1, \ldots, p.$

Note: α_i can be removed from inequality, and β_i can be removed from equality constraint. Minimizing $\alpha_0 f_0(x)$ is same as minimizing $f_0(x)$. Hence it is equivalent to

$$\begin{aligned} \text{minimize} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, i = 1, \ldots, m \\ & h_i(x) = 0, i = 1, \ldots, p \end{aligned}$$

Equivalent Problems: Change of Variables

Suppose $\phi : \mathbb{R}^n \to \mathbb{R}^n$ is one-one.

- Define $\tilde{f}_i$ as

$$\tilde{f}_i(z) = f_i(\phi(z)), \quad i = 0, \ldots, m$$

- Define $\tilde{h}_i$ as

$$\tilde{h}_i(z) = h_i(\phi(z)), \; i = 1, \ldots, p$$

Consider

$$\boxed{\begin{array}{rl} \text{minimize} & \tilde{f}_0(z) \\ \text{subject to} & \tilde{f}_i(z) \leq 0, \; i = 1, \ldots, m \\ & \tilde{h}_i(z) = 0, \; i = 1, \ldots, p \end{array}}$$

- If z solves above, then $x = \phi(z)$ solves the standard optimization problem.
- Similarly, if x solves original opt problem, then $z = \phi^{-1}(x)$ solves above

Equivalent Problems: Slack Variables

Given the optimization problem in standard form

Optimization problem (Standard Form):

$$\begin{aligned} \text{minimize} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, i = 1, \ldots, m \\ & h_i(x) = 0, i = 1, \ldots, p \end{aligned}$$

Quiz: Is it possible to replace inequality constraints by equality constraints and non-negativity constraints?
Ans: Yes. Key observation is: $f_i(x) \leq 0$, if and only if there is an $s_i \geq 0$ such that $f_i(x) + s_i = 0$

$$\begin{aligned} \text{minimize} \quad & f_0(x) \\ \text{subject to} \quad & s_i \geq 0, \quad i = 1, \ldots, m \\ & f_i(x) + s_i = 0, \ i = 1, \ldots, m \\ & h_i(x) = 0, \ i = 1, \ldots, p \end{aligned}$$

Note: Here s_i are called slack variable. Is this equivalent?

Convex Optimization Problem in Standard Form

Convex Optimization Problem (Standard Form):

$$\begin{aligned} \text{minimize} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & a_i^T x = b_i, \quad i = 1, \ldots, p \end{aligned}$$

where $f_0, \ldots, f_m$ are convex functions.

Comparing this with the standard form

Optimization problem (Standard Form):

$$\begin{aligned} \text{minimize} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, i = 1, \ldots, m \\ & h_i(x) = 0, i = 1, \ldots, p \end{aligned}$$

- objective function must be convex
- inequality constraint functions must be convex
- equality constraint functions $h_i(x) = a_i^T x - b_i$ must be affine

Convex Optimization Problem

Consider the following optimization problem with $x \in \mathbb{R}^2$

$$\begin{aligned} \text{maximize} \quad & f_0(x) = x_1^2 + x_2^2 \\ \text{subject to} \quad & f_1(x) = x_1/(1 + x_2^2) \leq 0 \\ & h_1(x) = (x_1 + x_2)^2 = 0, \end{aligned}$$

which is in standard form.
Quiz: Is this problem a convex optimization problem? Ans: No
Quiz: Can you rewrite this in convex optimization problem? Ans: Yes

$$\begin{aligned} \text{maximize} \quad & f_0(x) = x_1^2 + x_2^2 \\ \text{subject to} \quad & f_1(x) = x_1 \leq 0 \\ & h_i(x) = x_1 + x_2 = 0, \end{aligned}$$

Note: This is now a convex optimization problem

Outline

1. Optimization Model: Standard Form and Keywords
2. Optimality Conditions

Optimality Criteria for Constraint Optimization Problem

Optimality criteria

Suppose f_0 in a convex optimization problem is differentiable so that for all $x, y \in \text{dom } f_0$, we have

$$f_0(y) \geq f_0(x) + \nabla f_0(x)^T (y - x).$$

Let the feasible set X be

$$X = \{x \mid f_i(x) \leq 0, i = 1, \ldots, m, \quad h_i(x) = 0, i = 1, \ldots, p\}$$

Then x is optimal if and only if $x \in X$ and

$$\nabla f_0(x)^T (y - x) \geq 0 \quad \text{for all } y \in X.$$

Outline

1. Duality: Lagrange dual and KKT
   - Lagrangian dual function
   - Dual and conjugate functions
   - Weak Duality

Lagrangian Dual Function

Optimization problem (Standard Form):

$$\begin{aligned} \text{minimize} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, i = 1, \ldots, m \\ & h_i(x) = 0, i = 1, \ldots, p \end{aligned}$$

Idea: Augment the objective $f_0(x)$ with a weighted sum of the constraint functions.

Lagrangian: Define Lagrangian $L : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \to \mathbb{R}$ as

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x),$$

with dom $L = \mathcal{D} \times \mathbb{R}^m \times \mathbb{R}^p$. Here λ_i, ν_i are called Lagrange multipliers. Here λ and ν are called dual variables or Lagrange multiplier vectors.

Lagrange Dual Function

Lagrange Dual Function: Define the Lagrange dual function $g : \mathbb{R}^m \times \mathbb{R}^p \to \mathbb{R}$

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) = \inf_{x \in \mathcal{D}} \left(f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x) \right)$$

- If Lagrangian is unbounded below in x, the dual function takes on $-\infty$.
- The Lagrangian

$$L(x, \lambda, \nu) = \inf_{x \in \mathcal{D}} \left(f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x) \right)$$

  is affine as a function of λ and ν.
- Since $g(\lambda, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \nu)$, i.e., it is infimum of affine functions, $g(\lambda, \nu)$ is concave, **even when original problem is not convex!**

Lower Bounds on Optimal Value

Fact: The dual function yields lower bounds on the optimal value p^*. For any $\lambda \geq 0$ and any ν, we have

$$g(\lambda, \nu) \leq p^*.$$

Proof: Suppose $\tilde{x}$ is a feasible point, i.e., $f_i(\tilde{x}) \leq 0$ and $h_i(\tilde{x}) = 0$, and $\lambda \geq 0$. then

$$\sum_{i=1}^{m} \lambda_i f_i(\tilde{x}) + \sum_{i=1}^{p} \nu_i h_i(\tilde{x}) \leq 0$$

$$\implies L(\tilde{x}, \lambda, \nu) = f_0(\tilde{x}) + \sum_{i=1}^{m} \lambda_i f_i(\tilde{x}) + \sum_{i=1}^{p} \nu_i h_i(\tilde{x}) \leq f_0(\tilde{x})$$

$$\implies g(\lambda, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) \leq L(\tilde{x}, \lambda, \nu) \leq f_0(\tilde{x}), \quad \text{for any feasible } \tilde{x}$$

$$\implies g(\lambda, \nu) \leq \inf_{x \in \mathcal{D}} f_0(x) = p^*$$

Examples: Least Squares Solution of Linear Equation

Problem-1: Consider the following optimization problem

$$\begin{aligned} \text{minimize} \quad & x^T x \\ \text{subject to} \quad & Ax = b, \quad A \in \mathbb{R}^{p \times n} \end{aligned}$$

- Find the Lagrangian function
- Find the dual function
- Check whether the dual function is concave
- Check whether the dual function is lower bound to p^*

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Answer: On chalkboard!

Examples: Standard Form LP

Problem-2: Consider an LP in standard form

$$\begin{aligned} \text{minimize} \quad & c^T x \\ \text{subject to} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

- Find the Lagrangian
- Find the dual function
- Check whether the dual function gives lower bound to p^*

Two way Partitioning Problem

Problem-3:

$$\begin{array}{ll} \text{minimize} & x^T W x \\ \text{subject to} & x_i^2 = 1, \quad i = 1, \ldots, n, \end{array}$$

where $W \in S^n$.

- Find the Lagrangian
- Find the dual function
- Check whether dual is a lower bound for p^*

Lagrange Dual and Conjugate Functions-I

Recall that the conjugate f^* of a function $f : \mathbb{R}^n \to \mathbb{R}$ is given by

$$f^*(y) = \sup_{x \in \mathrm{dom}\ f} (y^T x - f(x)).$$

- Conjugate and Lagrange are closely related. Consider

$$\begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & x = 0 \end{array}$$

- Lagrangian is: $L(x, \nu) = f(x) + \nu^T x$, and dual function

$$g(\nu) = \inf_x (f(x) + \nu^T x) = -\sup_x ((-\nu)^T x - f(x)) = -f^*(-\nu)$$

- Note: Above, we used: $\inf_x f(x) = -\sup_x (-f(x))$

Lagrange and Conjugate Functions-II

Consider the following optimization problem

$$\begin{aligned} \text{minimize} \quad & f_0(x) \\ \text{subject to} \quad & Ax \leq b \\ & Cx = d. \end{aligned}$$

The dual is given as follows

$$\begin{aligned} g(\lambda, \nu) &= \inf_x (f_0(x) + \lambda^T (Ax - b) + \nu^T (Cx - d)) \\ &= -b^T \lambda - d^T \nu + \inf_x (f_0(x) + (A^T \lambda + C^T \nu)^T x) \\ &= -b^T \lambda - d^T \nu - f_0^*(-A^T \lambda - C^T \nu) \end{aligned}$$

- dom $g = \{(\lambda, \nu) \mid -A^T \lambda - C^T \nu \in \text{dom } f_0^*\}$

Equality Constraint Norm Minimization

Consider the problem

$$\begin{array}{ll} \text{minimize} & \|x\| \\ \text{subject to} & Ax = b \end{array}$$

where $\|\cdot\|$ is any norm. The conjugate of $f_0 = \|\cdot\|$ is given by

$$f_0^*(x) = \begin{cases} 0 & \|y\|_* \leq 1 \\ \infty & \text{otherwise} \end{cases}$$

We recall from above that

$$g(\lambda, \nu) = -b^T\lambda - d^T\nu - f_0^*(-A^T\lambda - C^T\nu)$$

Hence, for the given problem,

$$g(\nu) = -b^T\nu - f_0^*(-A^T\nu) = \begin{cases} -b^T\nu & \|A^T\nu\|_* \leq 1 \\ -\infty & \text{otherwise} \end{cases}$$

Lagrange Dual Problem

- For each $(\lambda, \nu), \lambda \geq 0$, Lagrange dual function gives lower bound for p^*
- The lower bound depends on λ, ν
- Question: what is the best lower bound?
- For best lower bound, solve the optimization problem

$$\begin{array}{ll} \text{maximize} & g(\lambda, \nu) \\ \text{subject to} & \lambda \geq 0 \end{array}$$

  This is called Lagrange dual problem
- Original optimization problem is called primal problem
- Pair (λ, ν), if $\lambda \geq 0$ and $g(\lambda, \nu) > -\infty$
- Optimal solution denoted by (λ^*, ν^*) is called dual optimal
- Optimization problem above is convex optimization problem
  - Regardless of whether primal problem is convex or not!

Weak Duality

- Let d^* denote the optimal value of Lagrange dual problem
- We have $d^* \leq p^*$, which holds even if the original problem is not convex

Weak duality: The condition

$$d^* \leq p^*$$

is called **weak duality**.

Optimal Duality Gap The difference $p^* - d^*$ is called optimal duality gap. The optimal duality gap is always nonnegative.

Strong Duality and Slater's condition

If $d^* = p^*$, then we say that strong duality holds.

Strong duality does not hold in general. But if the original problem is convex, then we usually have strong duality.

Slater condition: If there exists an $x \in \operatorname{int} \mathcal{D}$ such that

$$f_i(x) < 0, \quad i = 1, \ldots, m, \quad Ax = b.$$

then it is called Slater condition. Such a point x is called strictly feasible.

Slater's Theorem: Assume the primal problem is convex. If the Slater's condition holds, then the strong duality holds.

Slater's condition when some f_i are affine. There exists $x \in \operatorname{int} \mathcal{D}$ with

$$f_i(x) \leq 0, \quad i = 1, \ldots, k, \quad f_i(x) < 0, \quad i = k+1, \ldots, m, \quad Ax = b.$$

Outline

1. Duality: Lagrange dual and KKT
   - Weak Duality
   - Complementary Slackness
   - KKT Condition
   - Solving Primal via Dual
2. Duality Examples
3. Lecture-15: Optimization Models-I: Recommender Systems

Weak Duality

- Let d^* denote the optimal value of Lagrange dual problem
- We have $d^* \leq p^*$, which holds even if the original problem is not convex

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$$f_i(x) \leq 0, \quad i = 1, \ldots, k, \quad f_i(x) < 0, \quad i = k+1, \ldots, m, \quad Ax = b.$$

Example 1 (Strong Duality Example)

Recall the problem

$$\begin{aligned} \text{minimize} \quad & x^T x \\ \text{subject to} \quad & Ax = b. \end{aligned}$$

Quiz: What is the Slater's condition for this example?

Example 1 (Strong Duality Example)

Recall the problem

$$\begin{aligned} \text{minimize} \quad & x^T x \\ \text{subject to} \quad & Ax = b. \end{aligned}$$

Quiz: What is the Slater's condition for this example?

- There is no inequality to be satisfied
- Slater's condition: primal problem is feasible
- Case-1: $b \in \mathcal{R}(A)$, then $p^* = d^*$
- Case-2: $b \notin \mathcal{R}(A)$, then $p^* = -\infty$, but then $d^* = -\infty$

Hence, slater condition is satisfied always for this example. Hence strong duality holds always.

Problem: Lagrange dual and Slater's condition of QCQP

Consider the QCQP

$$\begin{array}{ll} \text{minimize} & (1/2)x^T P_0 x + q_0^T x + r_0 \\ \text{subject to} & (1/2)x^T P_i x + q_i^T x + r_i \leq 0, \quad i = 1, \ldots, m, \end{array}$$

with $P_0 \in S^n_{++}$, and $P_i \in S^n_+, i = 1, \ldots, m$.

Problem: Lagrange dual and Slater's condition of QCQP

Consider the QCQP

$$\begin{array}{ll} \text{minimize} & (1/2)x^T P_0 x + q_0^T x + r_0 \\ \text{subject to} & (1/2)x^T P_i x + q_i^T x + r_i \leq 0, \quad i = 1, \ldots, m, \end{array}$$

with $P_0 \in S^n_{++}$, and $P_i \in S^n_+, i = 1, \ldots, m$.

- Write the Slater's condition

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- Write the Lagrangian and dual function
- Write the dual problem

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- Write the Lagrangian and dual function
- Write the dual problem

Answer: See [page 227, Boyd]

Problem: Nonconvex Quadratic Problem with Strong Duality

Consider the following nonconvex problem

$$\begin{aligned} \text{minimize} \quad & x^T A x + 2b^T x \\ \text{subject to} \quad & x^T x \leq 1, \end{aligned}$$

$A \in S^n, A \not\succeq 0$, and $b \in \mathbb{R}^n$. Since $A \not\succeq 0$, this is not a convex problem.

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Problem: Nonconvex Quadratic Problem with Strong Duality

Consider the following nonconvex problem

$$\begin{aligned} \text{minimize} \quad & x^T A x + 2b^T x \\ \text{subject to} \quad & x^T x \leq 1, \end{aligned}$$

$A \in S^n, A \ngeq 0$, and $b \in \mathbb{R}^n$. Since $A \ngeq 0$, this is not a convex problem.

- What is the Lagrangian function and dual function?
- Write the dual problem.

Remark: Although, it is a nonconvex problem, strong duality holds. A slightly general condition for strong duality is:

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- What is the Lagrangian function and dual function?
- Write the dual problem.

Remark: Although, it is a nonconvex problem, strong duality holds. A slightly general condition for strong duality is:

General Strong Duality Condition:

Strong duality holds for any optimization problem with **quadratic** objective and one **quadratic inequality** constraint, provided Slater's condition holds. That is, there exists x feasible such that the quadratic inequality is **strict**.

Optimality Conditions

Certificate or Proof: If there exists a dual feasible point (λ, ν), then

$$p^* \geq g(\lambda, \nu)$$

Here we say that (λ, ν) provides a proof or certificate that above condition holds.

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Suboptimilaty of Primal Variable using Dual Variable: If x is primal feasible and (λ, ν) is dual feasible, then

$$f_0(x) - p^* \leq f_0(x) - g(\lambda, \nu)$$

That is x is $\epsilon-$suboptimal with $\epsilon = f_0(x) - g(\lambda, \nu)$.

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That is x is $\epsilon-$suboptimal with $\epsilon = f_0(x) - g(\lambda, \nu)$.

Duality gap associated with x and (λ, ν) : The gap

$$f_0(x) - g(\lambda, \nu)$$

is called the duality gap associated with x and (λ, ν).

Optimality Conditions

Certificates for Primal and Dual: If the duality gap corresponding to primal dual feasible pair x and (λ, ν) is zero, then $f_0(x) = g(\lambda, \nu)$, then x is primal optimal and (λ, ν) is dual optimal.
In this case,

- (λ, ν) is a certificate that proves that x is primal optimal
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In this case,

- (λ, ν) is a certificate that proves that x is primal optimal
- x is a certificate that proves that (λ, ν) is dual optimal

Non-heuristic Stopping Criteria If an algorithm produces a sequence of primal feasible x^k, and a sequence of dual feasible $(\lambda^{(k)}, \nu^{(k)}), k = 1, \ldots,$ and if $\epsilon_{\text{abs}} > 0$ is a given required accuracy. Then the stopping criteria

$$f_0(x^{(k)}) - g(\lambda^k, \nu^k) \leq \epsilon_{\text{abs}}$$

guarantees that the algorithm is $\epsilon-$suboptimal.

Complementary Slackness

Complementary Slackness: Suppose x^* is the primal optimal and (λ^*, ν^*) is the dual optimal, then

$$\begin{aligned}
f_0(x^*) &= g(\lambda^*, \nu^*) \qquad \text{(i.e., duality gap is zero)} \\
&= \inf_x \left(f_0(x) + \sum_{i=1}^m \lambda_i^* f_i(x) + \sum_{i=1}^p \nu_i^* h_i(x^*) \right) \qquad \text{(definition of dual)} \\
&\leq f_0(x^*) + \sum_{i=1}^m \lambda_i^* f_i(x^*) + \sum_{i=1}^p \nu_i^* h_i(x^*) \\
&\leq f_0(x^*) \qquad \text{(since, } \lambda_i^* \geq 0, f_i(x) \leq 0, h_i(x^*) = 0\text{)}
\end{aligned}$$

Conclusions:

- Inequality in 3rd line must be equality. Hence x^* minimizes $L(x, \lambda^*, \nu^*)$
- $\sum_{i=1}^m \lambda_i^* f_i(x^*) = 0$ because inequality in 4th line must be equality
- Hence

$$\lambda_i^* f_i(x) = 0, i = 1, \ldots, m.$$

This is called **complementary slackness**.

KKT conditions:

- Assume $f_0, f_1, \ldots, f_m, h_1, \ldots, h_p$ differentiable, no assumption on convexity.
- Let x^* and (λ^*, ν^*) be the primal and dual optimal with zero duality gap.
- Since x^* minimizes $L(x, \lambda^*, \nu^*)$ over x,

$$\nabla f_0(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla f_i(x^*) + \sum_{i=1}^{p} \nu_i^* \nabla h_i(x^*) = 0$$

- The KKT conditions are:

$$\begin{aligned} f_i(x^*) &\leq 0, \quad i = 1, \ldots, m \\ h_i(x^*) &= 0, \quad i = 1, \ldots, p \\ \lambda_i^* &\geq 0, \quad i = 1, \ldots, m \\ \lambda_i^* f_i(x^*) &= 0, \quad i = 1, \ldots, m \end{aligned}$$

$$\nabla f_0(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla f_i(x^*) + \sum_{i=1}^{p} \nu_i^* \nabla h_i(x^*) = 0$$

Fact: (Differentiable obj., constraint)+(Strong duality) $\implies$ KKT condition

KKT Conditions for Convex

Fact: When the primal problem is convex, then the KKT conditions are also sufficient for the points to be primal and dual optimal. That is, if f_i convex, and h_i affine, then for any $\tilde{x}, \tilde{\lambda}, \tilde{\nu}$ that satisfy KKT conditions

$$\begin{aligned} f_i(\tilde{x}) &\leq 0, \quad i = 1, \ldots, m \\ h_i(\tilde{x}) &= 0, \quad i = 1, \ldots, p \\ \tilde{\lambda}_i &\geq 0, \quad i = 1, \ldots, m \\ \tilde{\lambda}_i f_i(\tilde{x}) &= 0, \quad i = 1, \ldots, m \end{aligned}$$

$$\nabla f_0(\tilde{x}) + \sum_{i=1}^{m} \tilde{\lambda}_i \nabla f_i(\tilde{x}) + \sum_{i=1}^{p} \tilde{\nu}_i \nabla h_i(\tilde{x}) = 0,$$

then $\tilde{x}$ and $(\tilde{\lambda}, \tilde{\nu})$ are primal and dual optimal with zero duality gap.

Examples: KKT Conditions

Example-1: Consider the problem

$$\begin{array}{ll} \text{minimize} & (1/2)x^T P x + q^T x + r, \\ \text{subject to} & Ax = b, \end{array}$$

where $P \in S_+^n$. Write the KKT conditions for this problem.

Answer: On chalkboard.

Solving the primal problem via dual

Fact: Suppose strong duality holds. Suppose dual problem is solved and dual optimal (λ^*, ν^*) is known. Moreover, suppose the minimizer of $L(x, \lambda^*, \nu^*)$ exists, i.e., the solution of

$$\text{minimize} \quad f_0(x) + \sum_{i=1}^{m} \lambda_i^* f_i(x) + \sum_{i=1}^{p} \nu_i^* h_i(x) \tag{1}$$

is unique. If the solution of (1) is primal feasible, then it is primal optimal.

Outline

1. Duality: Lagrange dual and KKT
2. Duality Examples
3. Lecture-15: Optimization Models-I: Recommender Systems

KKT Examples

An optimization problem with equality constraint

Consider the problem

$$\begin{array}{rl} \text{minimize} & (1/2)x^T P x + q^T x + r \\ \text{subject to} & Ax = b, \end{array}$$

where $P \in S^n_+$. Write the KKT conditions.

Water-Filling Problem

Water-filling problem

Consider the convex optimization problem

$$\begin{array}{ll} \text{minimize} & -\sum_{i=1}^{n} \log(\alpha_i + x_i), \ \alpha_i > 0 \\ \text{subject to} & x \geq 0, \quad 1^T x = 1 \end{array}$$

1. Problem arises in information theory: allocating power to a set of n communication channels.
2. x_i : transmitter power allocated to ith channel
3. $\log(\alpha_i + x_i)$: capacity or communication rate of this channel
4. Problem: allocate a total power of one to the channels, in order to maximize the total communication rate

Write KKT conditions for this problem.

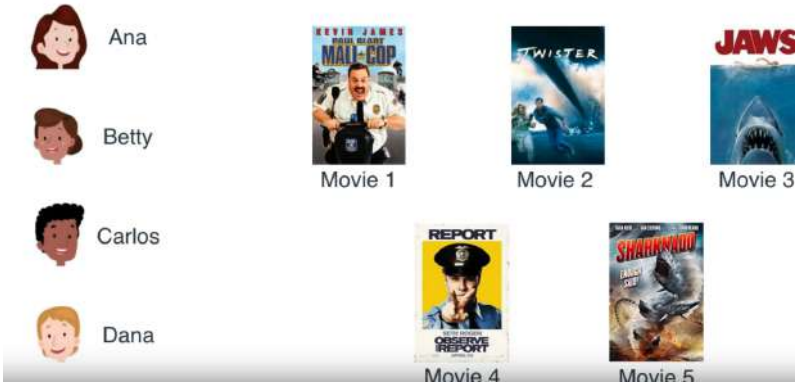
Outline

1. Duality: Lagrange dual and KKT
2. Duality Examples
3. Lecture-15: Optimization Models-I: Recommender Systems

Recommender Systems-I

Movie Recommendation: Given a history of movie ratings given by the user, recommend suitable movies that the user may like.

Example of Movie Ratings



- Consider five users and five movies
- Users are going to rate the movies from 1 to 5 stars
- Our goal is to predict ratings for the movie user has not rated yet

Example of Movie Ratings

Ana

Movie 5

	M1	M2	M3	M4	M5
					4

- For example, here Ana gives the rating 5 to movie M5
- Similarly, all other users will give their ratings for movies

Example of Movie Ratings

Ana

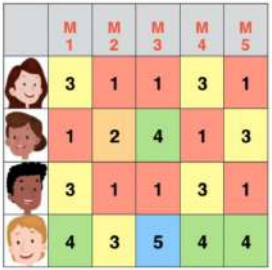
Movie 5

	M1	M2	M3	M4	M5
	1	3	2	5	4
	2	1	1	1	5
	3	2	3	1	5
	2	4	1	5	2

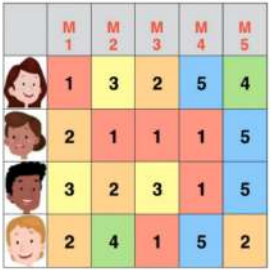
- When all users give their ratings for all the movies

Example of Movie Ratings

	M 1	M 2	M 3	M 4	M 5
	3	3	3	3	3
	3	3	3	3	3
	3	3	3	3	3
	3	3	3	3	3



	M 1	M 2	M 3	M 4	M 5
	3	1	1	3	1
	1	2	4	1	3
	3	1	1	3	1
	4	3	5	4	4



	M 1	M 2	M 3	M 4	M 5
	1	3	2	5	4
	2	1	1	1	5
	3	2	3	1	5
	2	4	1	5	2

- Which of the above movie ratings look realistic?
- The first one does not look realistic. Why?

Example of Movie Ratings

	M 1	M 2	M 3	M 4	M 5
	3	3	3	3	3
	3	3	3	3	3
	3	3	3	3	3
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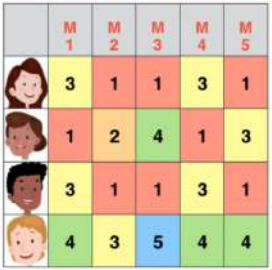

	M 1	M 2	M 3	M 4	M 5
	3	1	1	3	1
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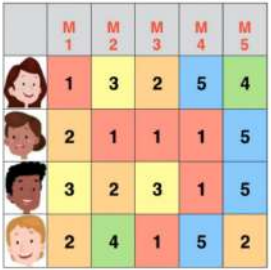
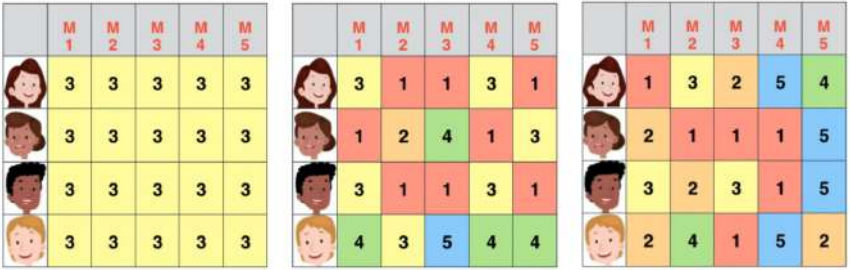


	M 1	M 2	M 3	M 4	M 5
	1	3	2	5	4
	2	1	1	1	5
	3	2	3	1	5
	2	4	1	5	2

- Which of the above movie ratings look realistic?
- The first one does not look realistic. Why?
  - Very rare that all users will give same ratings to all movies!

Example of Movie Ratings



	M 1	M 2	M 3	M 4	M 5
	3	3	3	3	3
	3	3	3	3	3
	3	3	3	3	3
	3	3	3	3	3

	M 1	M 2	M 3	M 4	M 5
	3	1	1	3	1
	1	2	4	1	3
	3	1	1	3	1
	4	3	5	4	4

	M 1	M 2	M 3	M 4	M 5
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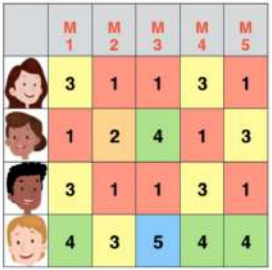
- Which of the above movie ratings look realistic?
- The first one does not look realistic. Why?
  - Very rare that all users will give same ratings to all movies!
- The rightmost one also does not look realistic. Why?

Example of Movie Ratings

	M 1	M 2	M 3	M 4	M 5
	3	3	3	3	3
	3	3	3	3	3
	3	3	3	3	3
	3	3	3	3	3



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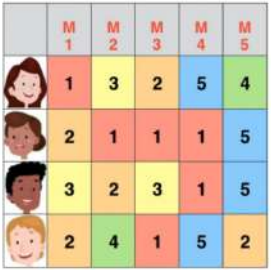
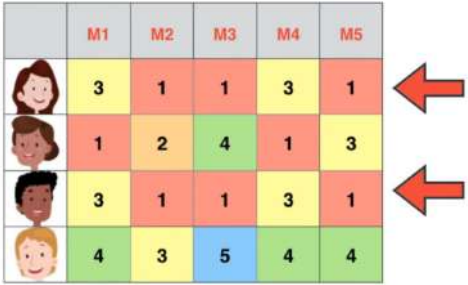


	M 1	M 2	M 3	M 4	M 5
	1	3	2	5	4
	2	1	1	1	5
	3	2	3	1	5
	2	4	1	5	2

- Which of the above movie ratings look realistic?
- The first one does not look realistic. Why?
  - Very rare that all users will give same ratings to all movies!
- The rightmost one also does not look realistic. Why?
  - When there are many movies and many users, it is very rare that all will give different ratings!
  - We expect dependencies!

Example of Movie Ratings



	M1	M2	M3	M4	M5
	3	1	1	3	1
	1	2	4	1	3
	3	1	1	3	1
	4	3	5	4	4

- Can you spot the dependencies in the rating matrix?
- Look at the 1st and 3rd row! They are same!

Example of Movie Ratings

	M1	M2	M3	M4	M5
	3	1	1	3	1
	1	2	4	1	3
	3	1	1	3	1
	4	3	5	4	4

- Can you spot other dependencies?

Example of Movie Ratings

	M1	M2	M3	M4	M5
	3	1	1	3	1
	1	2	4	1	3
	3	1	1	3	1
	4	3	5	4	4

- Can you spot other dependencies?
- Well, the 1st and 4th columns are same!

Example of Movie Ratings

	M1	M2	M3	M4	M5
	3	1	1	3	1
	1	2	4	1	3
	3	1	1	3	1
	4	3	5	4	4

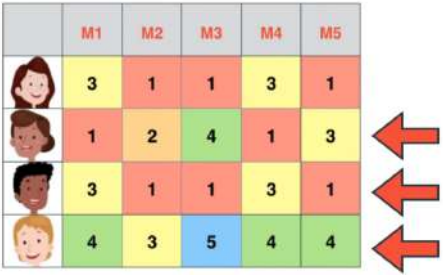
- Can you spot other dependencies?
- Well, the 1st and 4th columns are same!
- Does this imply movie M1 and M4 are same?

Example of Movie Ratings

	M1	M2	M3	M4	M5
	3	1	1	3	1
	1	2	4	1	3
	3	1	1	3	1
	4	3	5	4	4

- Can you spot other dependencies?
- Well, the 1st and 4th columns are same!
- Does this imply movie M1 and M4 are same?
- Well, not same but may be similar

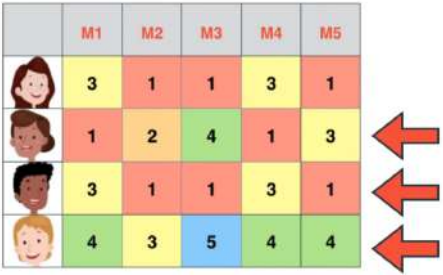
Example of Movie Ratings



	M1	M2	M3	M4	M5
	3	1	1	3	1
	1	2	4	1	3
	3	1	1	3	1
	4	3	5	4	4

- Can you find more dependencies?

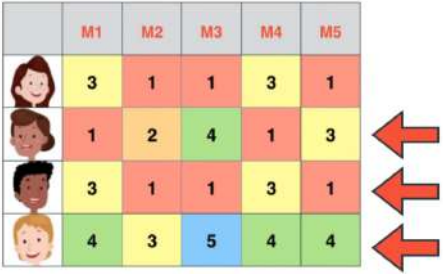
Example of Movie Ratings



	M1	M2	M3	M4	M5
	3	1	1	3	1
	1	2	4	1	3
	3	1	1	3	1
	4	3	5	4	4

- Can you find more dependencies?
- What are the dependencies of 3rd, 4th, and 5th row?

Example of Movie Ratings



	M1	M2	M3	M4	M5
	3	1	1	3	1
	1	2	4	1	3
	3	1	1	3	1
	4	3	5	4	4

- Can you find more dependencies?
- What are the dependencies of 3rd, 4th, and 5th row?
- Movie rating of the 4th user is sum of the ratings of user 3 and user 4.

Example of Movie Ratings

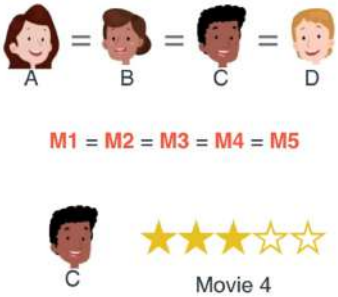
- Column 5 is average of column 2 and column 3

Example of Movie Ratings

- Column 5 is average of column 2 and column 3
- If M2 is a tornado movie, and M3 is shark movie, then M5 may be both shark and tornado movie

Guessing Ratings

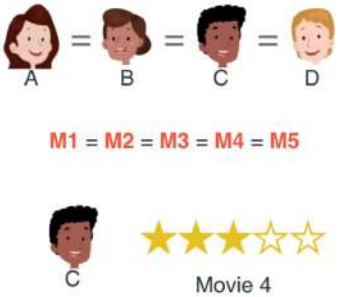
	M1	M2	M3	M4	M5
	3	3	3	3	3
	3	3	3	3	3
	3	3	3	?	3
	3	3	3	3	3



- User 3 did not watch movie M4

Guessing Ratings

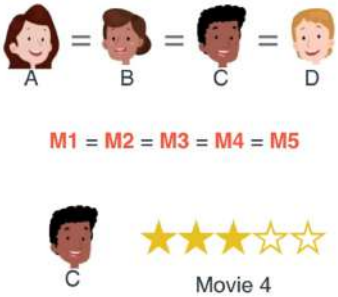
	M1	M2	M3	M4	M5
	3	3	3	3	3
	3	3	3	3	3
	3	3	3	?	3
	3	3	3	3	3



- User 3 did not watch movie M4
- Looking at the rating matrix, and dependencies, predict rating for user 3 for movie M4

Guessing Ratings

	M1	M2	M3	M4	M5
	3	3	3	3	3
	3	3	3	3	3
	3	3	3	?	3
	3	3	3	3	3



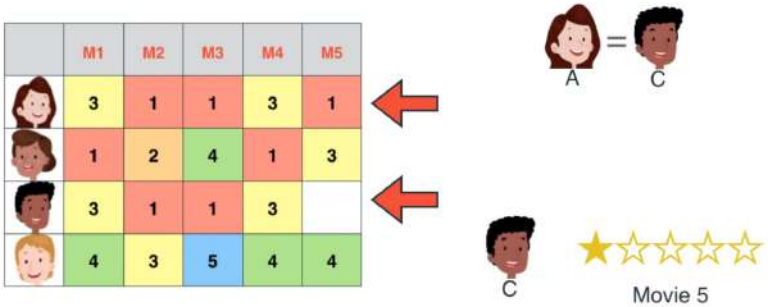
- User 3 did not watch movie M4
- Looking at the rating matrix, and dependencies, predict rating for user 3 for movie M4
- Well, we recommend rating 3

Guess Ratings

	M1	M2	M3	M4	M5
	3	1	1	3	1
	1	2	4	1	3
	3	1	1	3	
	4	3	5	4	4

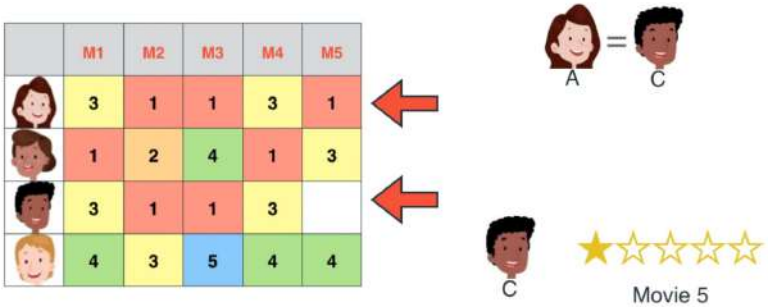
Quiz: Which rating to predict for user 3 for movie M5?

Guess Ratings



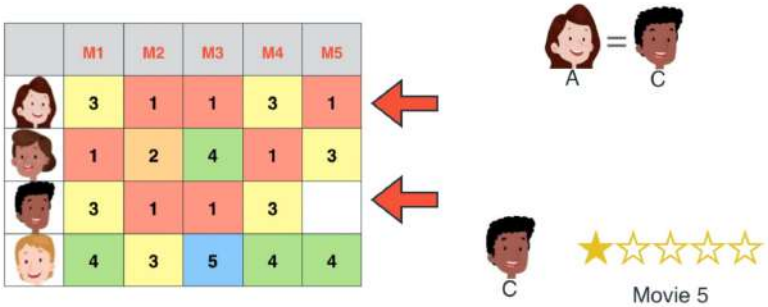
- Look at the ratings of Anna and Carlos!

Guess Ratings



- Look at the ratings of Anna and Carlos!
- They both have same ratings for all other movies!

Guess Ratings



- Look at the ratings of Anna and Carlos!
- They both have same ratings for all other movies!
- So, predict rating 1 for Carlos for M5 (same as Ana)

Guess Ratings

this × that =

	M1	M2	M3	M4	M5
	3	1	1	3	1
	1	2	4	1	3
	3	1	1	3	1
	4	3	5	4	4

- If the matrix has dependencies, it is low rank matrix

Guess Ratings

this × that =

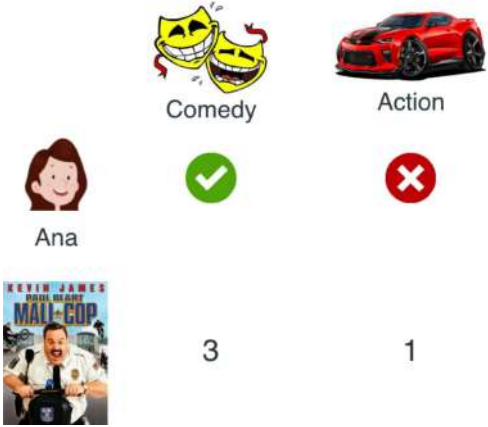
	M1	M2	M3	M4	M5
	3	1	1	3	1
	1	2	4	1	3
	3	1	1	3	1
	4	3	5	4	4

- If the matrix has dependencies, it is low rank matrix
- If matrix has rank r, then it can be written as product of rank r matrices
- How to do matrix factorization?

Guess Ratings

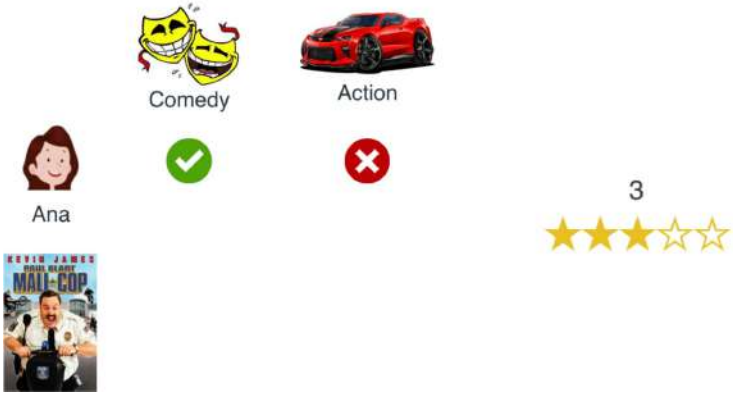
- Movies may have certain prominant features (corresponds to rank)
  - These are latent features. Ratings matrix does not tell explicitely about these features
- Most movies may be categorized into selected features

Guess Ratings



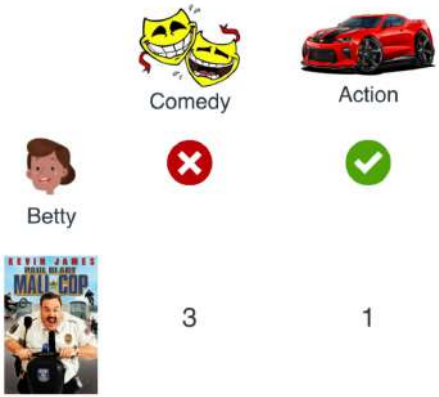
- Let there be two features
- Let Ana likes comedy and does not like action
- Also, let movie M1 has 3 of comedy and 1 of action

Guess Ratings

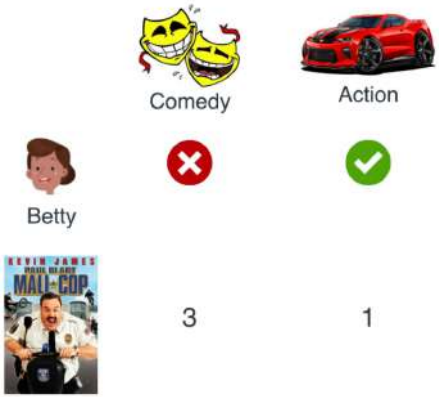


- Since Ana did not like action, Ana gives 3 rating:

$$3 \times \text{Comedy} + 0 \times \text{Action} = 3$$



- Betty does not likes Comedy and lieks Action



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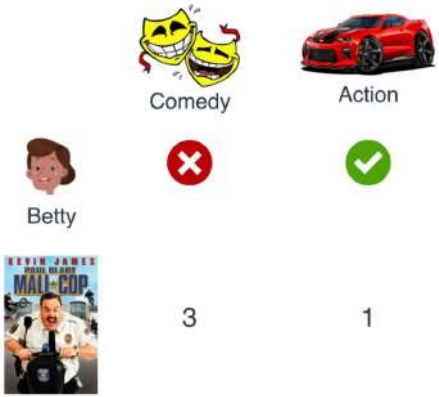

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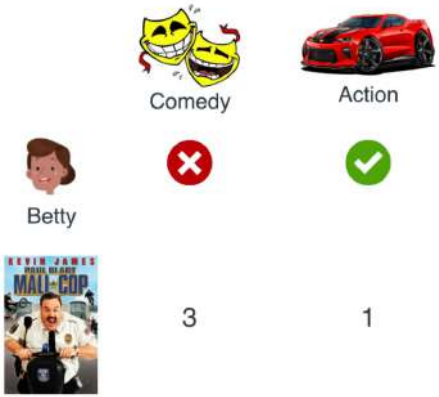
- Movie M1 has 3 of comedy and 1 of action
- Hence the rating given by Betty is 1 for movie 1:



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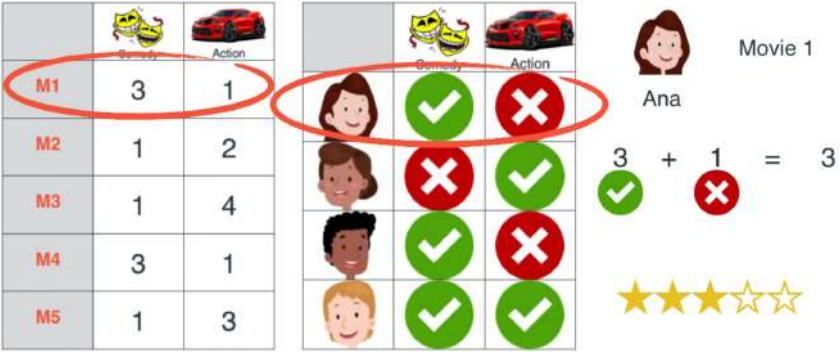
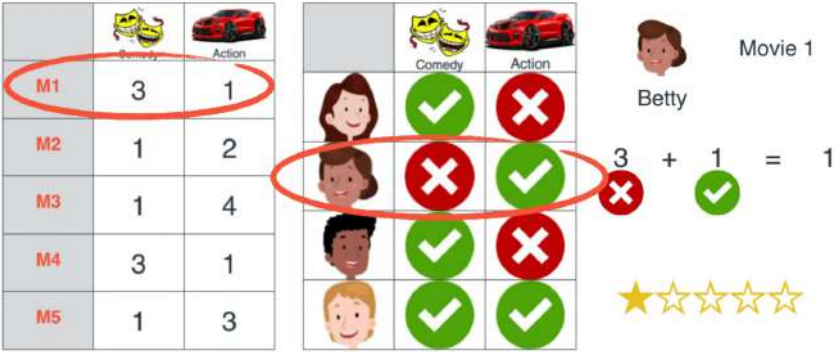
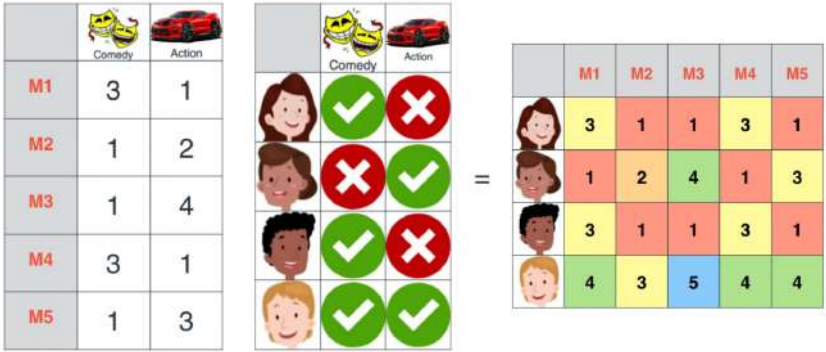
$$3 \times 0 + 1 \times 1 = 1$$



- Here we have combined movie features with features ratings for users



Betty gives the rating 1 for movie M1



(Movie feature matrix) $\times$ (User feature matrix) $=$ Ratings Matrix
Hence we have a matrix factorization!

Matrix Factorization

	M1	M2	M3	M4	M5
Comedy	3	1	1	3	1
Action	1	2	4	1	3

	Comedy	Action
A	✓	✗
B	✗	✓
C	✓	✗
D	✓	✓

	M1	M2	M3	M4	M5
	3	1	1	3	1
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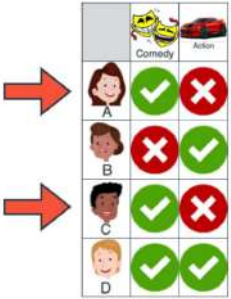
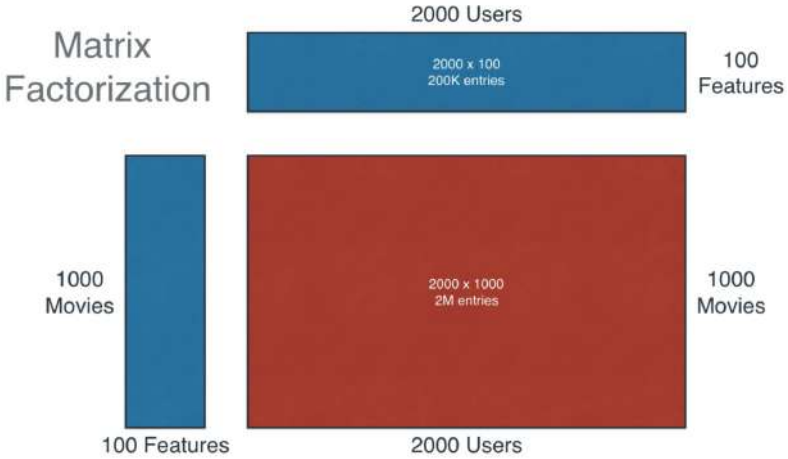


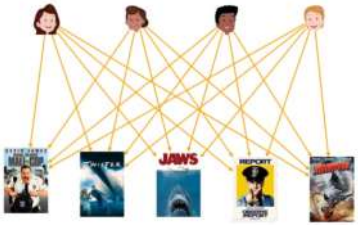
	M1	M2	M3	M4	M5
A	3	1	1	3	1
B	1	2	4	1	3
C	3	1	1	3	1
D	4	3	5	4	4

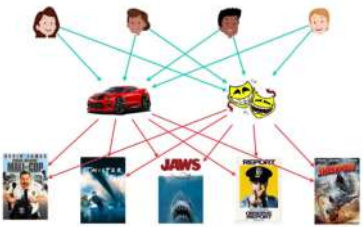
- Dependencies are preserved
- Row-1 and Row-2 are of ratings matrix are same
- Row-1 and Row-2 of user feature matrix are also same
- Similarly other dependencies



- Matrix factorization has benefits: storage!
- It is enough to store user feature matrix and movie feature matrix

Quiz: How many parameters (arrows)?





- Low rank factorization can be seen graphically

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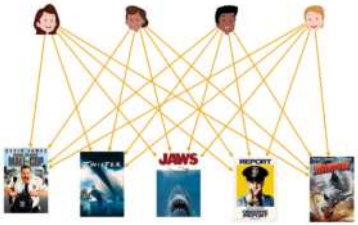

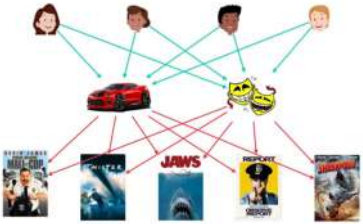

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- On the left, if we store the ratings of asll users for all movies then we need to store all edges

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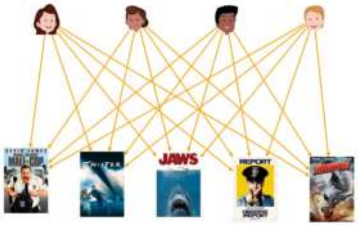
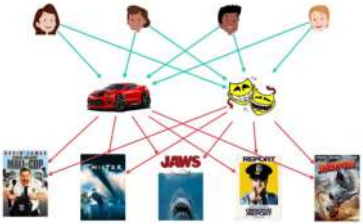
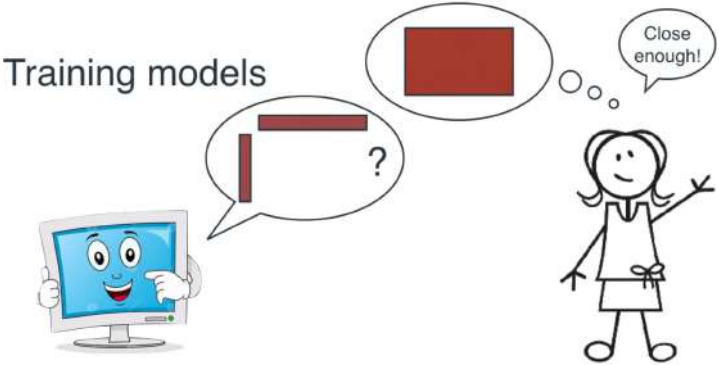




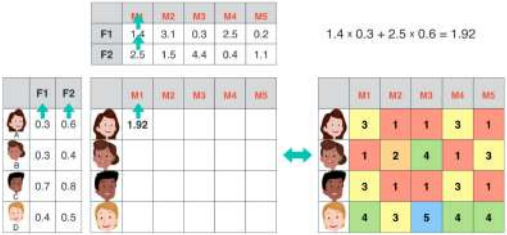
- Low rank factorization can be seen graphically
- On the left, if we store the ratings of asll users for all movies then we need to store all edges
- However, if we map user preference to features, then to movies, then less edges are needed.

Training the model



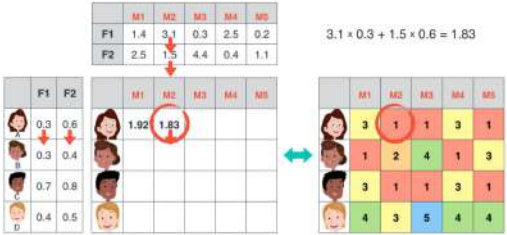
- Computer goes and comes with two factors, and compares to ground truth!
- If the match is not good, it keeps trying until it gets close

Predicted score of 1.44 does not match ground truth 3



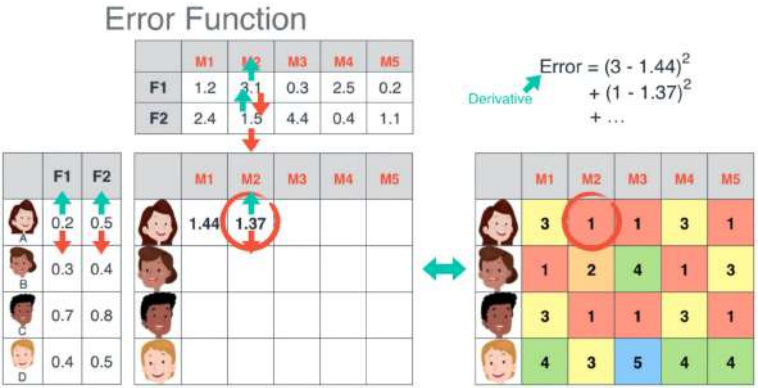
Increase the user feature rating and movie feature values

Here the predicted score 1.83 is more than the ground truth 1



Decrease the user feature rating and movie feature rating to make the predicted score closer to ground truth

Error Function



- Error function is defined as sum of the squares of the error in predicted scores and ground truth
- Gradient Methods can be used to minimize the error!

Four Ingredients in Supervised Machine Learning Models

Labelled Data: A data-label pair usually denoted by (X, Y) is needed.
Note: For recommender system: the data is rating matrix

Prediction Function: A function $f(W)$ that can predict or label a unseen data. Note: For recommender system: $f(W) = W_1 \cdot W_2$

Loss Function: A measure of how far the prediction function is from the given label.
Note: For recommender system, loss is $\|f(W) - Y\|_F^2$

Optimization Solve: A optimization solver to minimize the loss function as a function of W.
Note: For recommender systems, the optimization solver is gradient descent.

Optimization Model for Recommender System: Toy example

- Consider the following incomplete ratings matrix

$$A = \begin{bmatrix} 3 & & 1 \\ 1 & 1 & 4 \\ & 3 & 2 \end{bmatrix}$$

- Let the user feature matrix and movie feature matrix respectively be

$$P = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}, \quad Q^T = \begin{bmatrix} g & h \\ i & j \\ k & \ell \end{bmatrix}$$

- Let us create the indices for given ratings

$$K = \{(i,j) \mid A(i,j) \in A, \text{ that is, rating entry } (i,j) \text{ exists}\}$$

- Optimization Model:

$$\min_{P,Q}(\|PQ - A|_K\|_F), \quad A|_K \text{ is restriction of } A \text{ on set } K$$

Compute Loss Function for the Example

Given

$$P = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}, \quad Q = \begin{bmatrix} g & i & k \\ h & j & \ell \end{bmatrix}, \quad A = \begin{bmatrix} 3 & & 1 \\ 1 & 1 & 4 \\ & 3 & 2 \end{bmatrix}$$

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Computing PQ and comparing only with given entries of A above

$$ag + bh = 3, \quad ak + b\ell = 1, \quad cg + dh = 1, \quad ci + dj = 1$$
$$ck + d\ell = 4, \quad ei + fj = 3, \quad ek + f\ell = 2$$

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The loss is then defined by

$$\begin{aligned} \mathcal{L}(P, Q) = \|PQ - A_K\|_F^2 = (ag + gh - 3)^2 &+ (ak + b\ell - 1)^2 + (cg + dh - 1)^2 \\ &+ (ci + dj - 1)^2 + (ck + d\ell - 4)^2 \\ &+ (ei + fj - 3)^2 + (ek + f\ell - 2)^2 \end{aligned}$$

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Note: In particular, loss is a function of 12 variables: $a, b, c, d, \ldots$
Note: The loss function is non-convex. Why? (Hint: Second derivative test!)

Gradient Methods to Minimize Loss

Consider the loss function as before:

$$\begin{aligned}\mathcal{L}(a, b, \ldots, k, \ell) = {} & (ag + gh - 3)^2 + (ak + b\ell - 1)^2 + (cg + dh - 1)^2 \\ & + (ci + dj - 1)^2 + (ck + d\ell - 4)^2 + (ei + fj - 3)^2 + (ek + f\ell - 2)^2\end{aligned}$$

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To use gradient method, we need to compute gradients:

$$\begin{aligned}
\nabla_a \mathcal{L} &= 2(ag + gh - 3)g + 2(ak + b\ell - 1)k, \\
\nabla_b \mathcal{L} &= 2(ak + b\ell - 1)\ell \\
\nabla_c \mathcal{L} &= 2(cg + dh - 1)g + 2(ci + dj - 1)i + 2(ck + d\ell - 4)k \\
\nabla_d \mathcal{L} &= 2(cg + dh - 1)h + 2(ci + dj - 1)j + 2(ck + d\ell - 4)\ell \\
\nabla_e \mathcal{L} &= 2(ei + fj - 3)i + 2(ek + f\ell - 2)k, \\
\nabla_f \mathcal{L} &= 2(ei + fj - 3)j + 2(ek + f\ell - 2)\ell \\
\nabla_g \mathcal{L} &= 2(ag + gh - 3)(a + h) + 2(cg + dh - 1)c \\
\nabla_h \mathcal{L} &= 2(ag + gh - 3)g + 2(cg + dh - 1)d \\
&\vdots
\end{aligned}$$

Hence, $\nabla \mathcal{L} = (\nabla_a \mathcal{L}, \ldots, \nabla_\ell \mathcal{L})^T$

Gradient Method for Recommender Systems

[1] Loss function $\mathcal{L}, tol$ Initial weights: $W^0 = (a_0, b_0, \ldots, k_0, \ell_0)^T$ $\alpha = 1$ $c = 0 \ldots$
$W^{c+1} \leftarrow W^c - \alpha \nabla \mathcal{L}(W^c)$ $\|\nabla \mathcal{L}(W)\|_2 < tol$ exit $W = (a^*, b^*, \ldots, k^*, \ell^*)^T$, the optimal weights

Algorithm 1: Algorithm for training recommender system

Prediction for movie ratings

- Consider the ratings matrix with missing entries denoted by $\times$:

$$A = \begin{bmatrix} 3 & \times & 1 \\ 1 & 1 & 4 \\ \times & 3 & 2 \end{bmatrix}$$

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- Since we have now got optimal weights P^* and Q^* after gradient descent,

$$P^* \times Q^* = A = \begin{bmatrix} 3 & \times & 1 \\ 1 & 1 & 4 \\ \times & 3 & 2 \end{bmatrix}.$$

We can now predict the missing ratings by computing

$$A(1,2) = a^* i^* + b^* j^*$$
$$A(3,1) = e^* g^* + f^* h^*$$

Why did we not use SVD?

Consider extract from the paper: "Matrix Factorization Techniques for Recommender Systems", Y. Koren, R. Bell, C. Volinsky, IEEE, 2009

> Such a model is closely related to *singular value decomposition* (SVD), a well-established technique for identifying latent semantic factors in information retrieval. Applying SVD in the collaborative filtering domain requires factoring the user-item rating matrix. This often raises difficulties due to the high portion of missing values caused by sparseness in the user-item ratings matrix. Conventional SVD is undefined when knowledge about the matrix is incomplete. Moreover, carelessly addressing only the relatively few known entries is highly prone to overfitting.

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- Let $P, Q \in \mathbb{R}^{n \times r}$. Size of P and Q is hyperparameter

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$$p_u : \ u - th \text{ row of } P$$
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The $(u, i)th$ entry of A is given by $A(u, i)$

The regularized optimization model is given by:

$$\min_{P,Q} \sum_{(u,i) \in K} (A(u, i) - p_u q_i)^2 + \lambda(\|p\|^2 + \|q\|^2)$$

Here λ is the regularization parameter.

Dealing with Overfitting

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- Taking too big P and Q may lead to overfitting (large model!)(Why?)
- Penalize large P, Q by using regularization!
- Let us denote rows and columns of P, Q as follows

$$p_u : \ u - th \text{ row of } P$$
$$q_i : \ i - th \text{ column of } Q$$

The $(u, i)th$ entry of A is given by $A(u, i)$

The regularized optimization model is given by:

$$\min_{P,Q} \sum_{(u,i) \in K} (A(u, i) - p_u q_i)^2 + \lambda(\|p\|^2 + \|q\|^2)$$

Here λ is the regularization parameter.

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- Define the full gradient vector as follows

$$\nabla \mathcal{L}(P, Q) = \begin{bmatrix} \nabla_{p_1}(\mathcal{L}(p_1, q_1)) \\ \nabla_{p_2}(\mathcal{L}(p_2, q_1)) \\ \vdots \\ \nabla_{p_n}(\mathcal{L}(p_n, q_r)) \\ \vdots \\ \nabla_{q_1}(\mathcal{L}(p_1, q_1)) \\ \nabla_{q_2}(\mathcal{L}(p_1, q_2)) \\ \vdots \\ \nabla_{q_n}(\mathcal{L}(p_1, q_n)) \end{bmatrix}$$

Gradient Method for Regularized and Vectorized Recommender Systems

Regularized and Vectorized Recommender System is given below:

[1] Loss function $\mathcal{L}, tol$ Set initial random weights: $W^0 = (p_1^0, \ldots, p_n^0, q_1^0, \ldots, q_n^0)^T$ Set $\alpha = 1$ $c = 0 \ldots$ $W^{c+1} \leftarrow W^c - \alpha \nabla \mathcal{L}(W^c)$ $\|\nabla \mathcal{L}(W^c)\|_2 < tol$ exit $W^* = (p_1^*, \ldots, p_n^*, q_1^*, \ldots, q_n^*)^T$ the optimal weights

Algorithm 2: Training recommender system (Regularized+Vectorized)

Missing (i, j) entry is predicted by $p_i^* * q_j^*$